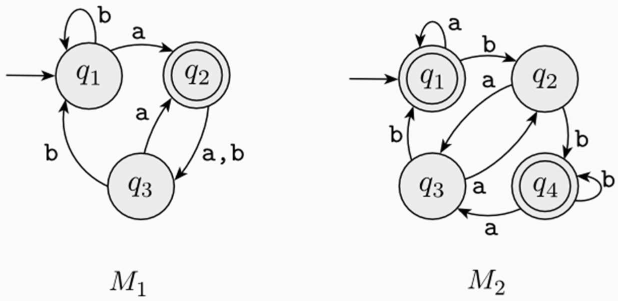
Assignment 1

Q1

1) [0.20 points]

The following are the state diagrams of two DFAs, M_1 and M_2. Answer the following questions about each of these machines.



a. What is the start state?
b. What is the set of accept states?
c. What sequence of states does the machine go through on input `aabb`?
d. Does the machine accept the string `aabb`?
e. Does the machine accept the string ε?

A1

(a)

For M_1, the start state is q_1. For M_2, the start state is q_1.

(b)

For M_1, $F = \{q_2\}$. For M_2, $F = \{q_1, q_4\}$.

(c)

For M_1, the sequence of states is q_1, q_2, q_3, q_1, q_1. For M_2, it is q_1, q_1, q_1, q_2, q_4.

(d)

M_1 will not accept but M_2 will.

(e)

M_1 will not accept but M_2 will.

Q2

2) [0.10 points]

Give the formal description of the two machines M_1 and M_2 pictured in exercise 1.

A2

$$M_1 = (\{q_1, q_2, q_3\}, \{\mathtt{a}, \mathtt{b}\}, \delta_1, q_1, \{q_2\})$$

$$M_2 = (\{q_1, q_2, q_3, q_4\}, \{\mathtt{a}, \mathtt{b}\}, \delta_2, q_1, \{q_1, q_4\})$$

δ_1	a	b
q_1	q_2	q_1
q_2	q_3	q_3
q_3	q_2	q_1

δ_2	a	b
q_1	q_1	q_2
q_2	q_3	q_4
q_3	q_2	q_1
q_4	q_3	q_4

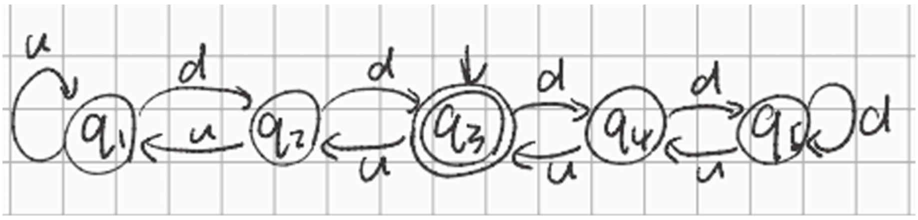
Q3

3) [0.10 points]

The formal description of a DFA M is $(\{q_1, q_2, q_3, q_4, q_5\}, \{\mathtt{u}, \mathtt{d}\}, \delta, q_3, \{q_3\})$, where δ is given by the following table. Give the state diagram of this machine.

	u	d
q_1	q_1	q_2
q_2	q_1	q_3
q_3	q_2	q_4
q_4	q_3	q_5
q_5	q_4	q_5

A3



Q4

4) [0.4 points]

Give the state diagrams of DFAs recognising the following languages over the alphabet $\Sigma = \{0,1\}$:

a) $\{w|w$ begins with a 0 and ends with a 1$\}$

b) $\{w|w$ contains at least four 1s$\}$

c) $\{w|w$ contains the substring 0101$\}$

d) $\{w|w$ has length at least 4 and its fourth symbol is a 1$\}$

e) $\{w|w$ starts with 1 and has odd length or starts with 0 and has even length$\}$

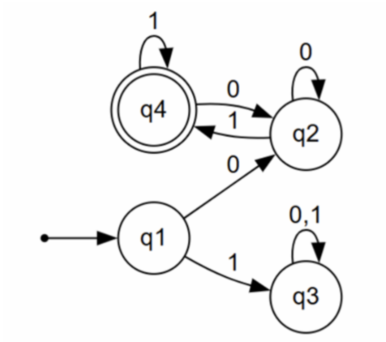
f) $\{w|w$ contains at least two 0s and at most one 1$\}$

g) The empty language

h) All strings except the empty string

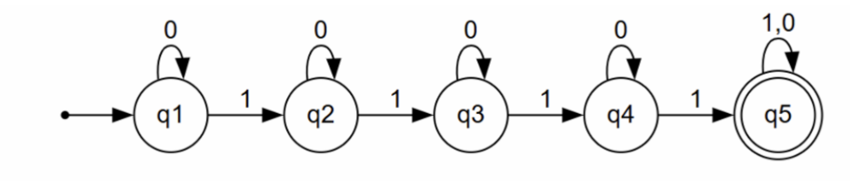
A4

a)



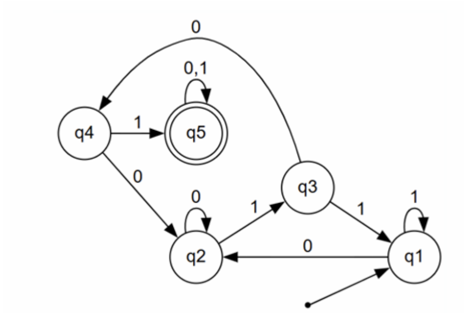
q1
q2
q3
q4
0
1
0,1

b)



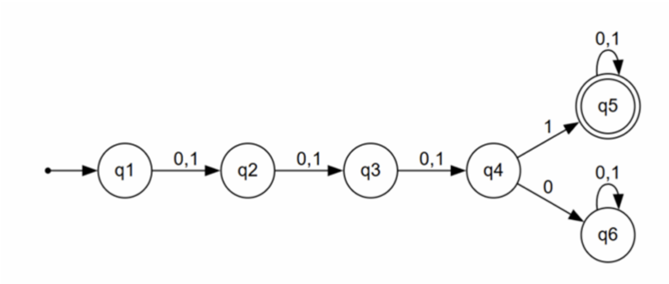
q1
q2
q3
q4
q5
0
1
1,0

c)



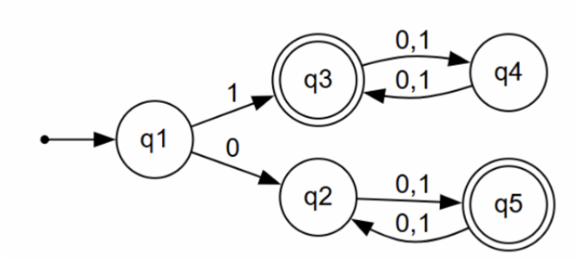
q1
q2
q3
q4
q5
0
1
0,1

d)



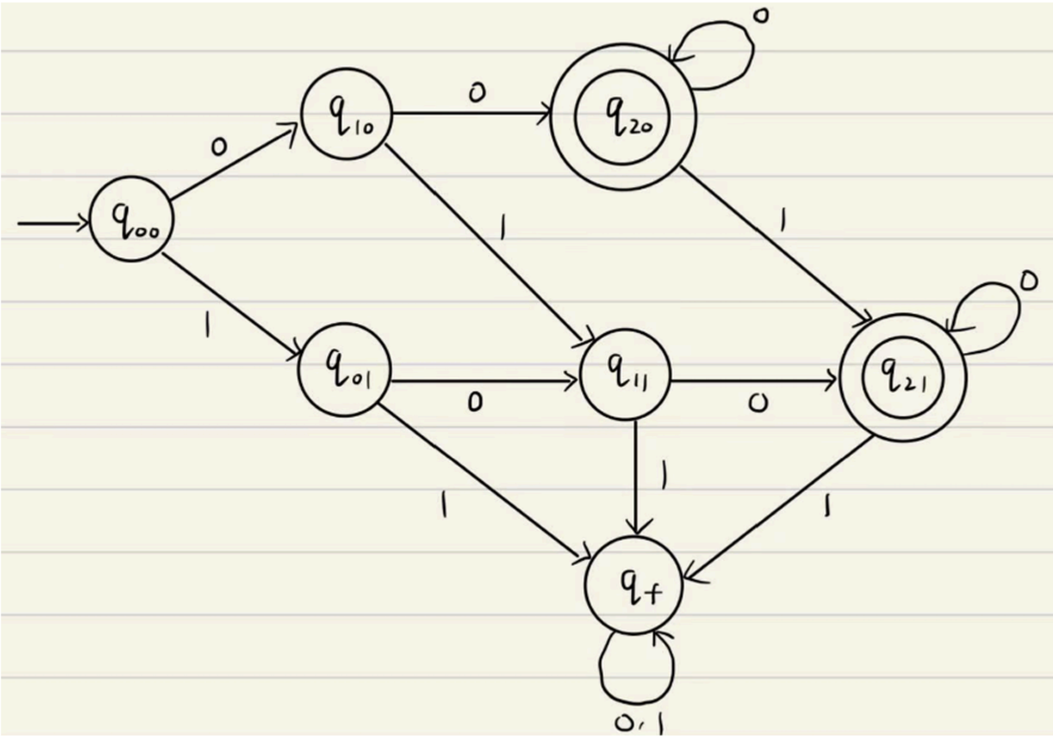
q1
q2
q3
q4
q5
q6
0,1
1
0

e)

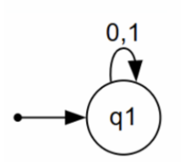


q1
q2
q3
q4
q5
1
0
0,1

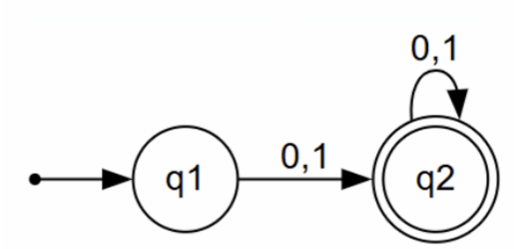
f)



g)



h)



Q5

5) [0.6 points]

Each of the following languages over $\Sigma = \{a, b\}$ is the complement of a simpler language. For each language construct a DFA for the simpler language and then use it to give the state diagram of a DFA for the original language.

a) $\{w | w$ does not contain the substring $ba\}$

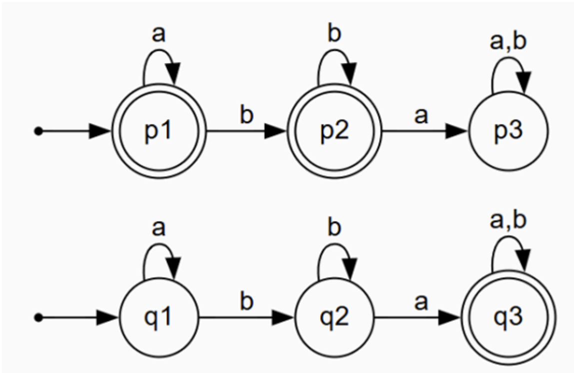
b) $\{w | w$ contains neither the substrings ab nor $ba\}$

c) $\{w | w$ is any string not in $b^*a^*\}$

A5

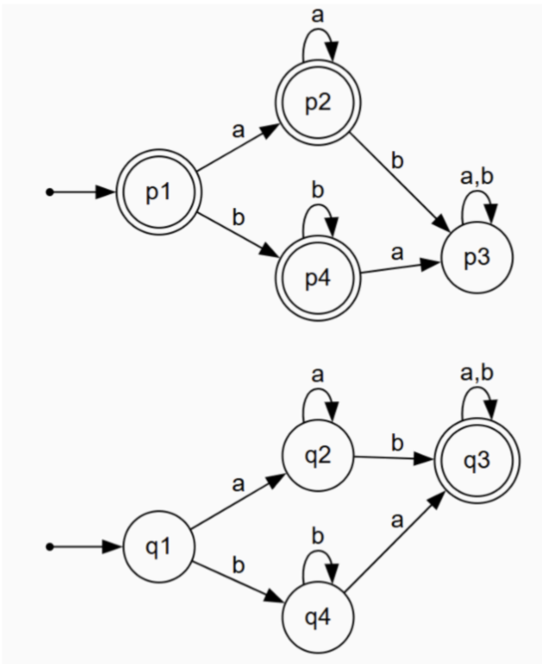
a)

$$\overline{L} = \{w \mid w \text{ does not contain the substring } \texttt{ba}\}$$
$$L = \{w \mid w \text{ contain the substring } \texttt{ba}\}$$



b)

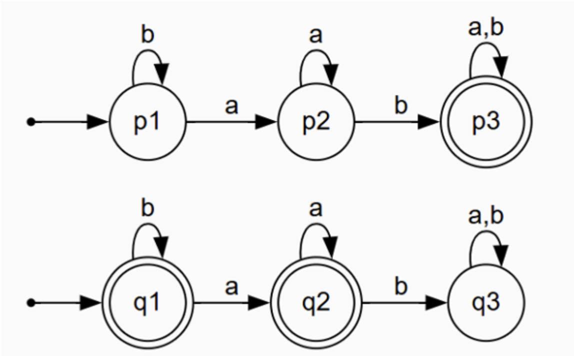
$$\overline{L} = \{w \mid w \text{ contains neither the substrings } \texttt{ab} \text{ nor } \texttt{ba}\}$$
$$L = \{w \mid w \text{ contains the substrings } \texttt{ab} \text{ or } \texttt{ba}\}$$



c)

$$\overline{L} = \{w \mid w \text{ is any string not in } \mathtt{b}^*\mathtt{a}^*\}$$

$$L = \{w \mid w \text{ is any string in } \mathtt{b}^*\mathtt{a}^*\}$$



Q6

6) [0.6 points]

Each of the following languages over $\Sigma = \{a, b\}$ is the intersection of two simpler languages. For each language construct DFAs for the simpler languages, then combine them using the intersection construction to give the state diagram of the original language.

a) $\{w|w$ has at least two a's and at least three b's$\}$

b) $\{w|w$ has an even number of a's and one or two b's$\}$

c) $\{w|w$ starts with an a and has at most two $b's\}$

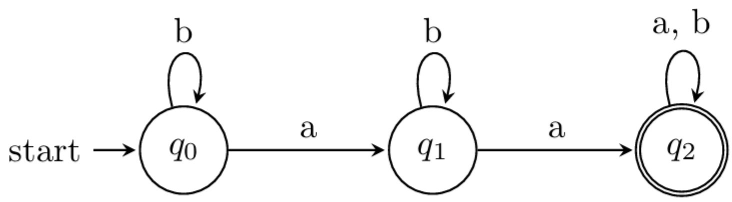
A6

a)

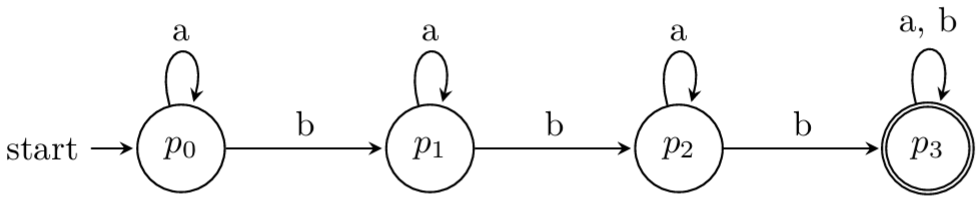
The simpler languages are:

- $L_1 = \{w \mid w \text{ has at least two a's}\}$
- $L_2 = \{w \mid w \text{ has at least three b's}\}$

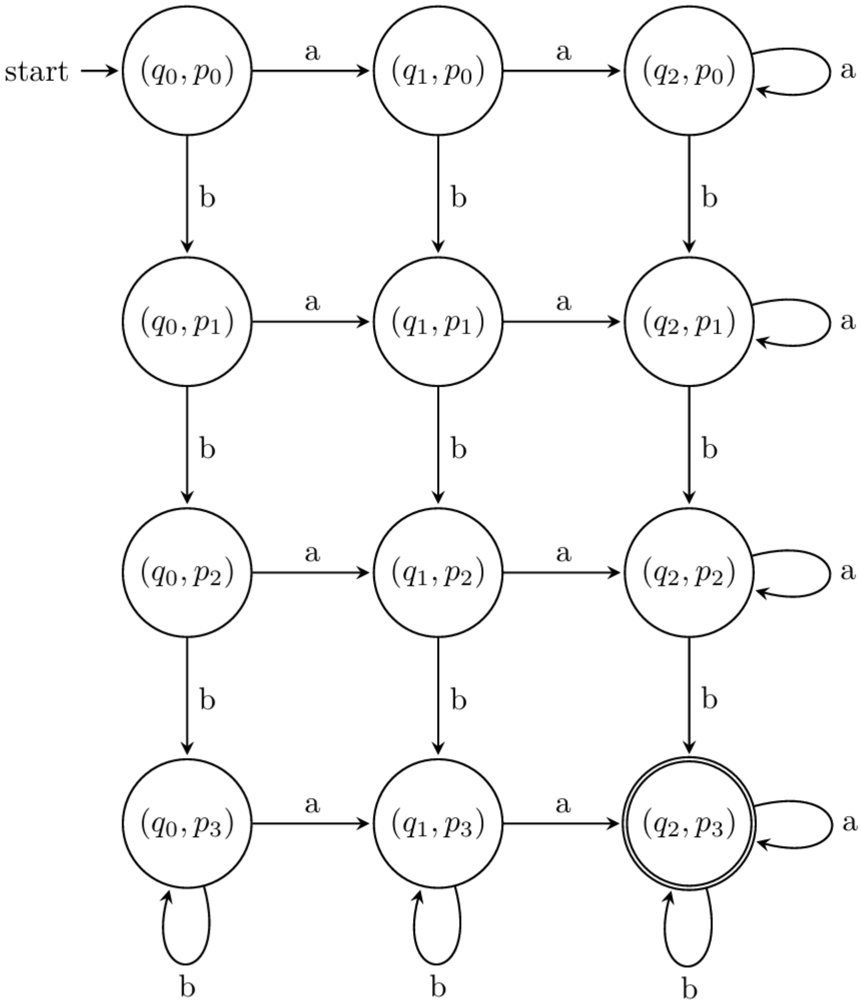
DFA for L_1 (at least two a's):



DFA for L_2 (at least three b's):



DFA for $L_1 \cap L_2$ (product construction, accepting states are (q_2, p_3)):

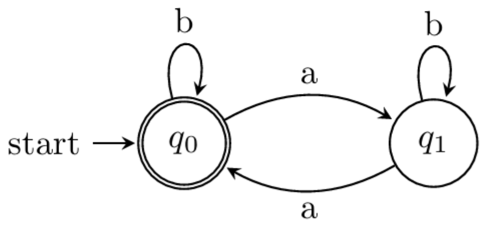


b)

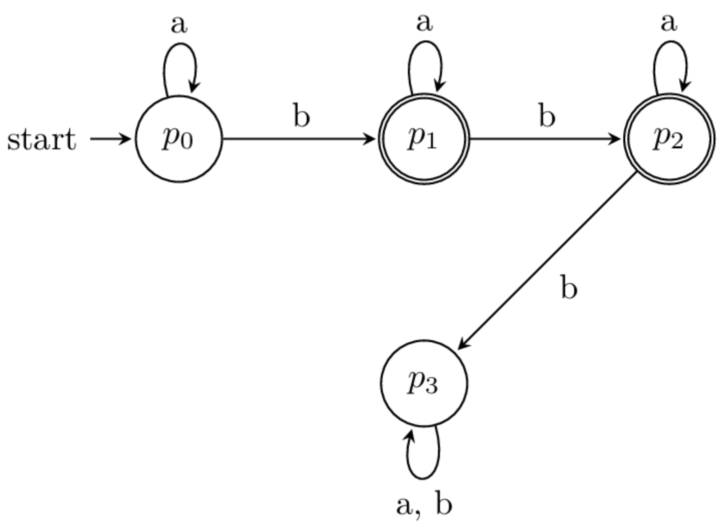
The two simpler languages are:

- $L_1 = \{w \mid w \text{ has an even number of a's}\}$
- $L_2 = \{w \mid w \text{ has one or two b's}\}$

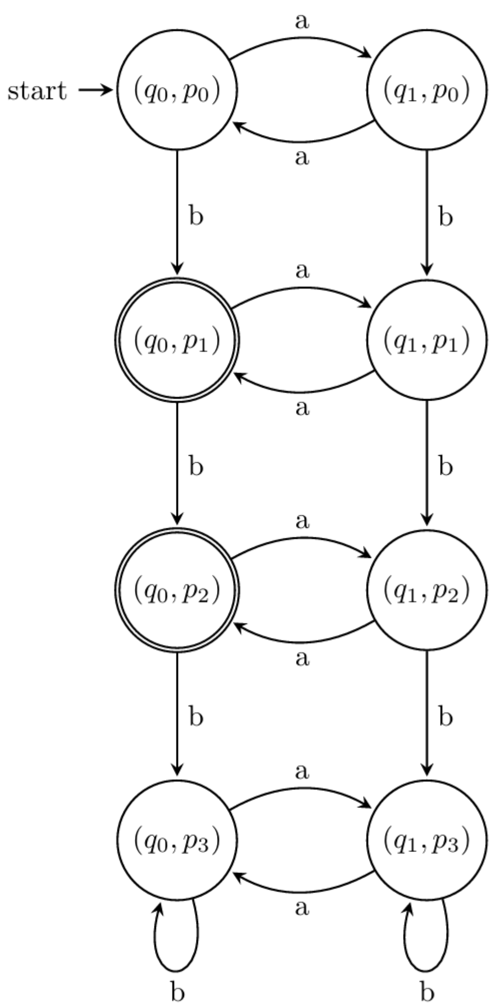
The state diagrams for L_1:



The state diagram for L_2:



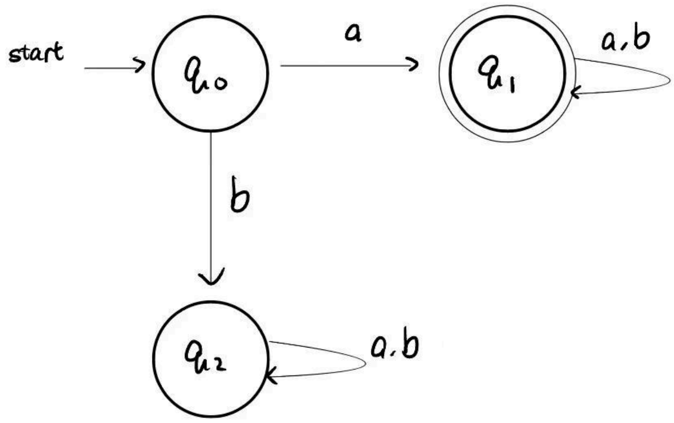
Thus the intersection DFA (accepting states are (q_0, p_1) and (q_0, p_2)):

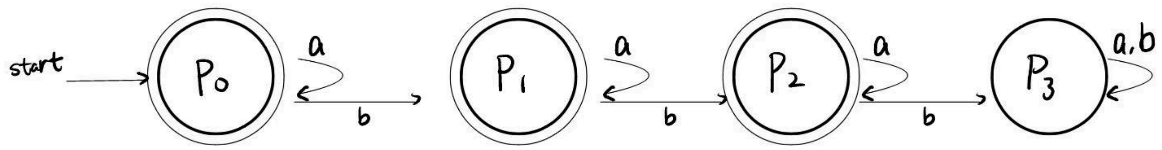


c)

The two simpler languages are:

- $L_1 = \{w \mid w \text{ starts with an a}\}$
- $L_2 = \{w \mid w \text{ has at most two b's}\}$





The intersection: